GAUTAM SINGH

Software Engineer, backend, data, and ML systems

Guelph, ON | (226) 979-6678 | gautamsingh2189@gmail.com | linkedin.com/in/gautam-singh-cs | github.com/Gautamstar | gautamstar.github.io/portfolio

EXPERIENCE

Machine Learning Intern Aug 2026 - Present

Sphere Global (automated vehicle damage inspection) *Remote*

- Build and maintain the Python tool the labeling team uses every day to prepare training data for an automated vehicle damage inspection system, running on one ordinary laptop with no GPU so customer photos never leave the site.
- Created a generator in NumPy and OpenCV that turns clean vehicle photos into realistic damaged ones, each labeled to the pixel in YOLO format, and it beat SDXL and FLUX on accuracy, speed, and cost for our use case.
- Found and fixed a flaw that had mislabeled 39% of training images, where the tool outlined the floor and walls instead of the car, by making a YOLO detector confirm the vehicle before SAM 2.1 traced it.

ServiceNow Developer & QA Specialist Sep 2024 - Dec 2024, Sep 2025 - Dec 2025

PAMT Consulting (now Ateko, a Bell Canada company) *Remote*

- Reduced new-hire onboarding handling time by ~30% by replacing a manual, error-prone process with an automated workflow built in ServiceNow App Engine Studio and Flow Designer.
- Designed and ran test cases across development, test, and production environments before each release.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Rust, Java, C++
Backend & Frontend: FastAPI, Django, Node.js, React, REST APIs, Redis/RQ, Flutter
Cloud & DevOps: AWS (Lambda, DynamoDB, CloudFront), Azure, Docker, Terraform, Linux, GitHub Actions CI/CD, Cloudflare
Data & ML: PostgreSQL, MongoDB, Redis, Databricks, PySpark, PyTorch, scikit-learn, Pandas, OpenCV, YOLO

PROJECTS

Distill *2026*

Python, PyTorch, Qwen2.5-VL, Transformers, Pydantic, FastAPI, Docker, GitHub Actions

- Raised accuracy from 80% to 97% for a local vision-language model that reads invoices and receipts into typed JSON, by sampling the model several times to score confidence per field and sending low-confidence fields to a person.
- Wrote 175 tests and a CI check that fails the build if any published accuracy number changes.

FitFileSize Live | GitHub | *2026*

Python, FastAPI, Redis/RQ, React, TypeScript, Docker, Cloudflare Tunnel, Ghostscript, MCP

- Solo-built and launched a web app, public API, and MCP server that compress PDFs and images (7 formats, including iPhone HEIC) to fit an upload limit or a form's exact rules, such as visa photos and exam uploads.
- Designed an async job pipeline (FastAPI, Redis/RQ workers, Server-Sent Events) with format-specific logic behind one strategy interface, self-hosted with Docker on a VPS behind a Cloudflare Tunnel with deploy-on-merge.
- Halved the work per file with a search that predicts the right compression setting from sizes already measured, taking a 5 MB photo from 9.2 s to 4.2 s in production with the same output, covered by ~350 tests across three CI jobs.
- Ran failure drills on the deployment and fixed what they found: a watchdog now restarts any container that fails its health check, and deploys give workers two minutes to finish jobs instead of Docker's default ten seconds.

Cairn Live | GitHub | *2026*

Rust, AWS Lambda, DynamoDB, CloudFront, API Gateway, Terraform

- Cut warm response to 5.1 ms and cold starts to 184 ms with a Rust binary on ARM64 AWS Lambda that does one DynamoDB write per event and computes everything else later.
- Deployed the full stack of Lambda, DynamoDB, CloudFront, and API Gateway as 41 Terraform managed resources running live across three sites at zero steady-state cost and made the service cookie-free by hashing visitor identifiers under a key that rotates daily.

The Almanac (Capstone) GitHub | *2025*

Python, Azure Data Factory, Databricks, PySpark, Azure SQL, PyTorch, Django REST, Flutter

- Improved data load performance by ~35% by redesigning the storage layer of an automated market-data pipeline behind a stock-tracking web and mobile app, shipped with a team over eight months.

OPEN SOURCE

actualbudget/actual PR #8603 | *Merged 2026*

- Contributed a merged bug fix to a widely used open-source personal finance app, returning a proper error when a password change is attempted with no password method set.

EDUCATION

Bachelor of Computer Science (Honors) Sep 2020 - Dec 2025

Conestoga College *Waterloo, ON*

Google Advanced Data Analytics Professional Certificate *Jun 2026*